

American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Computer Graphics [A]

Final-Term Project Report

PROJECT TITLE:

Dreamland Village Ultimate

An Interactive 2D Animated Village Simulation Using OpenGL & C++

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1 Introduction

1.1 Project Overview

Dreamland Village Ultimate is an interactive 2D animated village made with C++, OpenGL and the FreeGLUT toolkit. It shows a village of Bangladesh with houses, trees, a paddy field, animals, a river and a boat. No images or textures are used. Everything is drawn from basic shapes: points, lines, triangles, quads and polygons, inside one 2D coordinate space.

The project has four scene modes: **Day**, **Sunset**, **Night** and **Rain**. The user can switch between them at any time with the keyboard. Each mode changes the sky colours, the sun or moon, the effects (stars, fireflies, rain, lightning) and the background sound, while the village animation keeps running.

1.2 Objectives

- Practise drawing basic shapes, circles and ellipses, and using transformations (translate, rotate, scale).
- Draw a Bangladeshi village with cottages, banyan, palm and banana trees, a paddy field and a river.
- Use simple math (sine waves) to animate the cow, birds, trees, water, smoke, clothes and the boat.
- Make a rain effect with 260 falling drops, splashes on the ground and random lightning.
- Keep the animation smooth with a timer that updates the scene about 60 times per second.

1.3 Tools & Technologies Used

Component	Specification / Purpose
Programming Language	C++ (C++11)
Graphics API	OpenGL for drawing, GLU for the 2D projection (`gluOrtho2D`)
Window / Event Toolkit	FreeGLUT 3.0.0 for MinGW (included in the repository): window, keyboard, mouse, timer and bitmap text
Audio	Windows Multimedia (`winmm`, `PlaySound`) for `.wav` playback
Math	C++ `<cmath>` (`sin`, `cos`, `fabs`, `fmod`)
Compiler & IDE	MinGW GCC with Code::Blocks (Windows 64-bit)
Version Control	Git and GitHub

2 Project Proposal

2.1 Problem Statement

A village scene has many things moving at the same time: light changes during the day, water waves, trees moving in the wind and animals walking. Drawing all of this in pure 2D without any images needs math for the shapes and the motion, the right drawing order, and an animation loop that stays smooth.

2.2 Proposed Solution

We split the project into five parts: basic shapes and sound, landscape, animals and details, river and weather, and main control. Each part is a set of drawing functions. The `display()` function calls them from back to front, and a timer function updates all the moving values every frame.

3 Schematic Diagram & Coordinate System

3.1 Coordinate System Setup

The program uses a 2D orthographic projection, `gluOrtho2D(0, 100, 0, 80)`. So the world is 100×80 units, with the origin $(0, 0)$ at the bottom-left corner. The window is 1280×720 pixels and OpenGL scales the world to fit it.

3.2 Layered Rendering Hierarchy

There is no depth buffer, so objects drawn later appear in front (painter's algorithm). `display()` draws in this order:

1. **Sky:** sky gradient, sun or moon, rotating sun rays, stars, clouds and birds ($Y \in [38, 80]$).
2. **Land:** hills, paddy field, chicken, grass and the cow ($Y \in [17, 52]$).
3. **Village:** path, palm trees, both cottages, chimney and smoke, fences, banyan tree, banana plants, flower garden and clothesline ($Y \in [17, 62]$).
4. **River:** river with waves, ripples, sun/moon reflection, fish and the bridge ($Y \in [0, 20]$).
5. **Boat & Ducks:** rowing boat with the fisherman, and ducks ($Y \in [5, 16]$).
6. **Effects on Top:** fireflies, rain, lightning, rain splashes and the help panel.

4 List of Objects

ID	Object Name	Base Primitives	Visual Description
OBJ-01	Sky	GL_QUADS	Two-colour gradient that changes with the scene mode.
OBJ-02	Sun & Rays	GL_POLYGON, GL_LINES	Yellow sun circle with 12 rays that slowly rotate.
OBJ-03	Moon & Stars	GL_POLYGON, GL_POINTS	Crescent made from two circles; stars as points and small twinkling circles.
OBJ-04	Clouds	GL_POLYGON, GL_QUADS	Groups of white (grey in rain) ellipses moving slowly to the right.
OBJ-05	Birds	GL_LINE_STRIP	V-shaped birds crossing the sky with flapping wings.
OBJ-06	Hills & Field	GL_TRIANGLES, GL_POLYGON, GL_LINES	Hills, green land, paddy field and swaying grass.
OBJ-07	Cottages	GL_QUADS, GL_POLYGON	Two houses with roofs, doors and windows; windows glow at night.
OBJ-08	Trees	GL_LINE_STRIP, GL_POLYGON	Banyan, palm and banana trees whose leaves sway in the wind.
OBJ-09	Cow	GL_POLYGON, GL_TRIANGLES, GL_QUADS, GL_LINES	Cow walking left and right with moving legs, nodding head and swinging tail.
OBJ-10	River	GL_QUAD_STRIP, GL_LINES, GL_LINE_STRIP	Blue river with a moving sine-wave top edge and ripples.
OBJ-11	Boat	GL_POLYGON, GL_LINES	Wooden boat with a fisherman rowing an oar; can be steered.
OBJ-12	Rain	GL_LINES, GL_LINE_STRIP	260 slanted rain drops, ground splashes and random lightning.

5 Functions to Represent the Objects

The code is divided into five parts, one for each group member:

5.1 Part 1: Low-Level Primitives & Base Engine (Iftekhar Rahman)

- `rect(x1, y1, x2, y2)`: draws a filled rectangle with `GL_QUADS` (walls, legs, posts).
- `circle(cx, cy, r, parts)`: draws a filled circle with `GL_POLYGON`, using 70 points by default.
- `ellipse(cx, cy, rx, ry, parts)`: same as `circle` but with different X and Y radius.
- `line(x1, y1, x2, y2, width)`: draws a line with the given width.
- `drawText(x, y, text, font)`: writes text with `glutBitmapCharacter`.
- **Sound Functions**: `updateSound()` plays `nature.wav` (Day/Sunset), `night.wav` (Night) or `rain.wav` (Rain); `cowSoundTimer()` plays `cow.wav` every 10 seconds in Day and Sunset; `stopSound()` stops the sound.
- `drawIntro()`: start screen with the title, a subtitle, "PRESS ENTER TO START" and the list of modes.

5.2 Part 2: Village Landscape & Environment (Najiat Islam Rishad)

- `drawSky()`, `drawSunMoonStars()`, `drawSunRays()`, `drawTwinklingStars()`: sky colours, sun, moon, rotating sun rays and twinkling stars.
- `drawCloud()`, `drawClouds()`, `drawBird()`, `drawBirds()`: moving clouds and flying birds.
- `drawHills()`, `drawPaddyField()`, `drawGrassDetails()`, `drawVillagePath()`: hills, paddy field, grass and the village path.
- `drawPalmTree()`, `drawBanyanTree()`, `drawBananaPlant()`: trees with leaves swaying in the wind.
- `drawHouseOne()`, `drawHouseTwo()`, `drawChimney()`, `drawSmoke()`: two cottages, the chimney and rising smoke.

5.3 Part 3: Animals & Village Details (Faysal Talukder)

- `drawCow()`: cow that walks left and right; the legs move, the head nods and the tail swings.
- `drawChicken()`: chicken walking slowly across the yard with a small up-down step.
- `drawFence()`, `drawFlower()`, `drawFlowerGarden()`, `drawClothesline()`: fences, swaying flowers and clothes moving on the line.

5.4 Part 4: River, Boat & Weather (Md. Sazzadur Rahman)

- `drawRiver()`, `drawRiverRipples()`, `drawReflection()`, `drawBridge()`: river with waves, ripples, sun or moon reflection and the bridge.
- `drawBoatBody()`, `drawFisherman()`, `drawBoat()`: boat with a fisherman rowing; the boat can face left or right.
- `drawDuck()`, `drawDucks()`, `drawOneFish()`, `drawFish()`: ducks on the water; fish swim across the river and jump out of the water in Rain mode.
- `drawFireflies()`, `drawRain()`, `drawRainSplashes()`: fireflies at night, falling rain with lightning, and rain splashes.

5.5 Part 5: Main Control & Animation (Md. Arafat Rahman Samir)

- `display()`: draws the whole scene from back to front.
- `update()`: timer function that runs every 16 ms and changes all the moving values.
- `toggleFullScreen()`: switches between fullscreen and a 1280×720 window.
- `keyboard()`, `specialKeys()`, `mouse()`: handle keys, arrow keys and mouse clicks.
- `init()`, `main()`: create the window, set the projection and start the main loop.

6 Algorithms & Mathematical Techniques Used

6.1 Parametric Generation of Circles and Ellipses

A circle or ellipse is drawn as a polygon with $N = 70$ points. Point i is at angle $\theta_i = \frac{2\pi i}{N}$:

$$x_i = c_x + r_x \cdot \cos(\theta_i), \qquad y_i = c_y + r_y \cdot \sin(\theta_i) \qquad \text{for } i \in [0, N-1] \tag{1}$$

Setting $r_x = r_y$ gives a circle; different values give ellipses for clouds, the cow body, fish and ducks.

6.2 Harmonic Motion with Sine Functions

Most movement uses a sine wave. A phase value grows a little every frame, and the position or angle follows it:

$$y(t) = y_0 + A \cdot \sin(\text{phase}) \tag{2}$$

- **River Waves:** the top edge of the river is $y = 16 + 0.65 \cdot \sin(0.2x + \text{riverPhase})$.
- **Cow Legs:** the legs shift by $0.28 \cdot \sin(\text{cowLegPhase})$, so front and back legs move in opposite directions.
- **Tree Sway:** palm leaves rotate by $2.8 \cdot \sin(\text{palmPhase} + 0.1x)$ degrees, so each tree moves a little differently.

6.3 Matrix Transformations & Pivot Rotations

To rotate an object around its own point (pivot), it is moved to the origin, rotated, and moved back. `glPushMatrix()` and `glPopMatrix()` keep this change only for that object:

$$P' = T(x_{\text{pivot}}, y_{\text{pivot}}) \cdot R(\theta) \cdot T(-x_{\text{pivot}}, -y_{\text{pivot}}) \cdot P \tag{3}$$

This is used for the sun rays, the swaying trees, the cow's head and the fisherman's oar.

7 Interactive Functions & System Control

Part 5 handles the keyboard, mouse, window mode and frame timing.

7.1 Input Command Reference

Input Trigger	System Response
`Enter`	Closes the intro screen and starts the village.
`1 / D`	**Day Mode**: blue sky, sun with rotating rays, plays `nature.wav`.
`2`	**Sunset Mode**: purple-orange sky with a low red sun, plays `nature.wav`.
`3 / N`	**Night Mode**: dark sky, moon, stars, fireflies, plays `night.wav`.
`4 / R`	**Rain Mode**: grey sky, rain, lightning, jumping fish, plays `rain.wav`.
`B / b`	Starts or stops the boat.
`L / l`	Turns the house window lights on or off (seen at night).
`H / h`	Shows or hides the help panel.
`F / f`	Switches fullscreen on or off.
`M / U`	Turns the background sound off / on.
`Left / Right Arrow`	Boat moves to the left / right.
`Mouse Left / Right`	Boat moves to the right / left.
`Mouse Middle`	Stops the boat.
`Esc`	Stops the sound and closes the program.

7.2 Frame Pipeline & Timer Dispatcher

The animation runs with a GLUT timer about 60 times per second:

```
glutTimerFunc(16, update, 0);
```

Each call moves the clouds, birds, boat, cow and other objects, updates the rain drops, sometimes starts a lightning flash, then calls `glutPostRedisplay()` to draw the new frame.

8 Scenario Descriptions & Output Screenshots

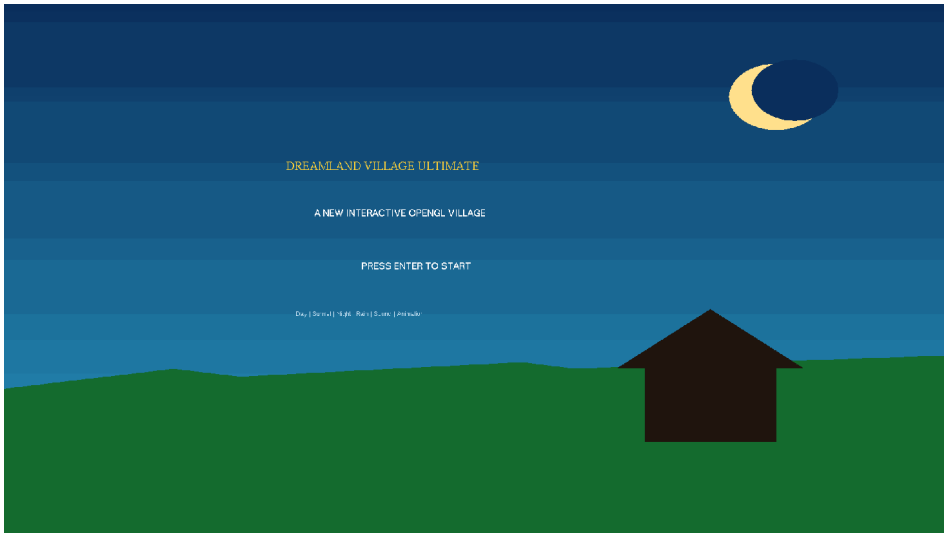


Figure 1: **Introductory Scene:** title, subtitle and "PRESS ENTER TO START".

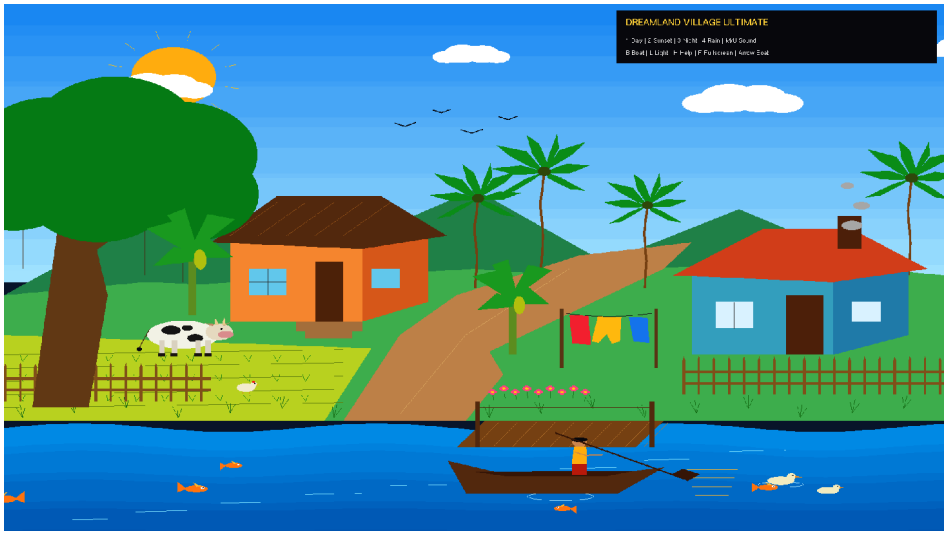


Figure 2: **Daytime Village Scene:** blue sky, sun with rays, paddy field, cottages, walking cow and the boat.



Figure 3: **Sunset Mode:** purple-orange sky and a red sun going down.



Figure 4: **Night Simulation:** dark sky, twinkling stars, crescent moon, fireflies and lit cottage windows.



Figure 5: **Rain Scenario:** grey sky, slanted rain, splashes and jumping fish, with rain sound.

9 Limitations & Future Improvements

9.1 Current Limitations

- **Fixed Motion:** the boat and animals follow fixed paths; there is no collision between objects.
- **Single-File Code:** all code is in one file (`main.cpp`) instead of separate classes or files.
- **Windows-Only Sound:** sound uses the Windows `PlaySound` function (`winmm`), so it works only on Windows.
- **Fixed World Size:** the world is a fixed 100×80 area, so on a screen with a different shape the fullscreen picture is stretched.

9.2 Future Enhancements

- Use a cross-platform sound library such as OpenAL or SDL2_mixer.

- Move the scene to 3D with a camera and lighting.
- Let villagers and animals move around more freely.
- Add more seasons, for example kash flowers in autumn and morning fog in winter.

10 Conclusion

Dreamland Village Ultimate shows the main ideas of 2D computer graphics: drawing basic shapes, circles and ellipses, using transformations, painter's algorithm and timer-based animation. The village runs smoothly at about 60 frames per second and changes between Day, Sunset, Night and Rain with matching sounds.

11 Source Code & Repository

The complete source code, FreeGLUT files, sounds and screenshots are available at:

`https://github.com/munshi-rishad/ComputerGraphics-Dreamland-Village-Ultimate`

To run it, open `openglportable-64bit/openglportable/openglportable.cbp` in Code::Blocks and press **F9**, then press **Enter** on the intro screen.